



2026

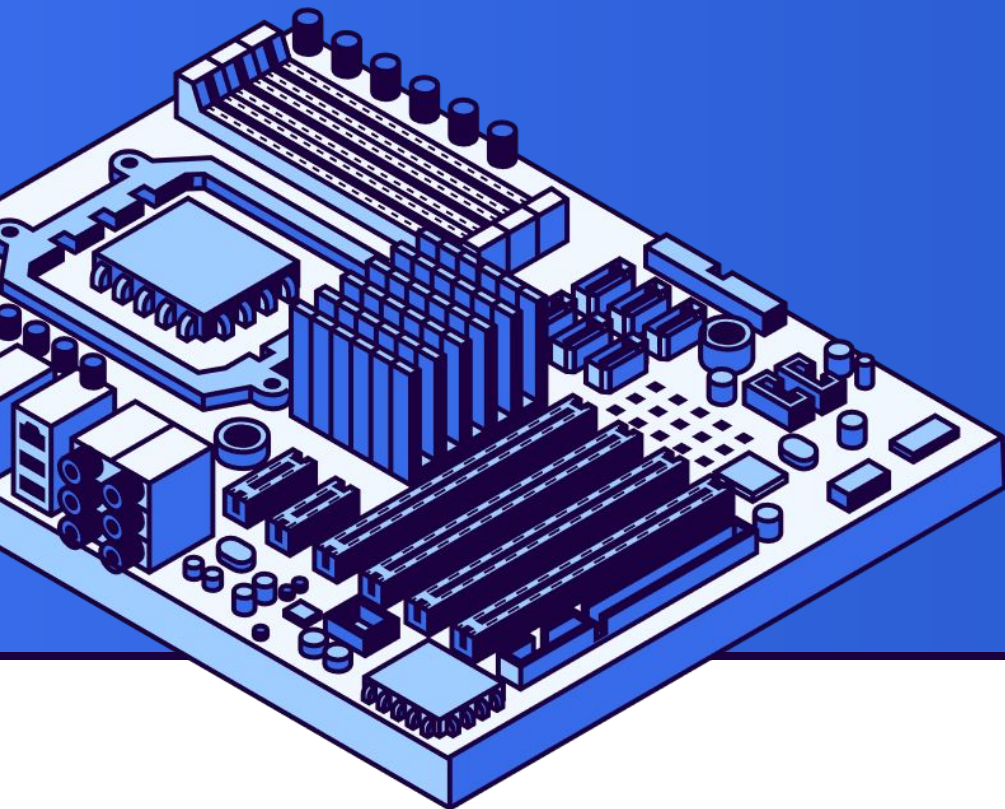
Operating System For Banking Transaction Processing



CSA0401-OPERATING SYSTEMS

Presented By

srinivas s - 192524129





INTRODUCTION

- Banks process thousands of transactions every second
- N**• The OS manages CPU, memory, storage, and networking.
- Ensures secure, reliable, and fast transaction processing.
- Supports multiple users simultaneously with high availability.
- Example: ATM withdrawals, UPI transfers, online banking,





Requirements of Banking Operating System

- High Performance
- High Reliability (24×7 operation)
- Strong Security
- Fault Tolerance
- Scalability
- Fast Response Time
- Data Integrity





Types of Operating Systems

Suitable

MULTIUSER OPERATING SYSTEM

Supports thousands of users simultaneously.

MULTIPROCESSING OPERATING SYSTEM

Uses multiple CPUs for faster processing.

DISTRIBUTED OPERATING SYSTEM

Connects multiple bank servers across branches.

REAL-TIME OPERATING SYSTEM

Handles time-sensitive banking operations.



OS Structure

HYBRID/LAYERED KERNEL STRUCTURE
Used
Layers.

- User Applications
- Banking Software
- System Call
Interface
- Kernel



Role of System Calls in

SYSTEM CALLS ALLOW BANKING APPLICATIONS TO COMMUNICATE WITH THE OS

Examples:

Banking

File Management

Process Creation

Memory Allocation

Network Communication

Device Management

Security & Authentication

Transaction Flow

Processing

Customer Request



Authentication



System Call



Database Access



Transaction Processing



Balance Update





Process Management

Operating System performs:

Multitasking

Process Scheduling

Thread Management

CPU Allocation

Deadlock Prevention



Advantage

S

- High Speed Processing
- Secure Transactions
- 24×7 Availability
- Efficient Resource Utilization
- Supports Millions of Users
- Prevents Data Loss
- Reliable Banking Operations



Conclusion

- Banking systems require powerful and secure operating systems.
- Multiuser, Multiprocessing, Distributed, and Soft Real-Time OS are most suitable.
- Hybrid kernel architecture provides performance and security.
- System calls enable communication between banking applications and the operating system, ensuring safe and efficient transaction processing.





Thank You